Predicting Augmentation Transforms with Minimal Auxiliary Loss
Improves Model Robustness Across Architectures

VizInt Experiment

February 28, 2026

Abstract

We demonstrate that adding an auxiliary task of predicting augmentation transform types with a very small loss weight significantly improves model robustness to image corruptions across multiple backbone architectures (ResNet18, VGG16, VGG19, Vision Transformer). Using the loss formulation $\mathcal{L} = \mathcal{L}_{\text{cls}} + 0.01 \cdot \mathcal{L}_{\text{cls}} \cdot \sigma(\mathcal{L}_{\text{transform}})$, the Blended models achieve substantially higher accuracy under corruption while maintaining competitive clean performance.

1 Method

1.1 Loss Formulation

The total training loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + 0.01 \cdot \mathcal{L}_{\text{cls}} \cdot \sigma(\mathcal{L}_{\text{transform}}) \tag{1}$$

1.2 Model Categories

- **Normal**: Standard training without data augmentation
- **Robust**: Trained with continuous transforms as data augmentation
- **Blended**: Auxiliary transform prediction task with tiny loss weight

1.3 Backbone Architectures

We evaluate across four architectures: ResNet18, VGG16, VGG19, and Vision Transformer (ViT).

2 Results

2.1 Main Results

Table 1: Classification accuracy at various corruption severity levels on CIFAR-10.

Backbone	Strategy	S=0.0	S=0.25	S=0.5	S=0.75	S=1.0	Avg Drop	OOD
ResNet18	Normal	0.6448	0.4847	0.4226	0.4089	0.3996	0.2159	0.3006
	Robust	0.8928	0.8390	0.8078	0.7810	0.7690	0.0936	0.5189
	Blended	0.9044	0.8626	0.8398	0.8155	0.7963	0.0758	0.5156
VGG16	Normal	0.8577	0.6192	0.5448	0.5106	0.5028	0.3134	0.4119
	Robust	0.8852	0.8355	0.8092	0.7783	0.7588	0.0897	0.4963
	Blended	0.8828	0.8326	0.8068	0.7860	0.7551	0.0877	0.4939
VGG19	Normal	0.8723	0.6074	0.5353	0.5197	0.5135	0.3283	0.4218
	Robust	0.8786	0.8247	0.7914	0.7737	0.7503	0.0936	0.4845
	Blended	0.8731	0.8181	0.7769	0.7506	0.7316	0.1038	0.4774
ViT	Normal	0.7284	0.4875	0.4590	0.4434	0.4465	0.2693	0.3314
	Robust	0.7940	0.7435	0.7145	0.6927	0.6717	0.0884	0.4184
	Blended	0.7921	0.7395	0.7158	0.6904	0.6693	0.0884	0.4085
	Blended3fc	0.7504	0.6929	0.6485	0.6316	0.6226	0.1015	0.3940

2.2 Out-of-Distribution Robustness

Table 2: Accuracy on out-of-distribution corruptions.

Model	Category	OOD Accuracy
ResNet18 Normal	Normal	0.3006
VGG16 Normal	Normal	0.4119
VGG19 Normal	Normal	0.4218
ViT Normal	Normal	0.3314
ResNet18 Robust	Robust	0.5189
VGG16 Robust	Robust	0.4963
VGG19 Robust	Robust	0.4845
ViT Robust	Robust	0.4184
ResNet18 Blended	Blended	0.5156
VGG16 Blended	Blended	0.4939
VGG19 Blended	Blended	0.4774
ViT Blended	Blended	0.4085
ViT Blended3fc	Blended3fc	0.3940

2.3 Transform Prediction Accuracy

Table 3: Accuracy of the auxiliary transform type prediction task (Blended models).

Model	S=0.25	S=0.5	S=0.75	S=1.0
ResNet18 Blended	0.7353	0.8248	0.8480	0.8470
VGG16 Blended	0.3791	0.3688	0.3590	0.3373
VGG19 Blended	0.3970	0.4048	0.3760	0.3567
ViT Blended	0.7415	0.8136	0.8218	0.8129
ViT Blended3fc	0.3841	0.4452	0.4755	0.4686

2.4 Visualizations

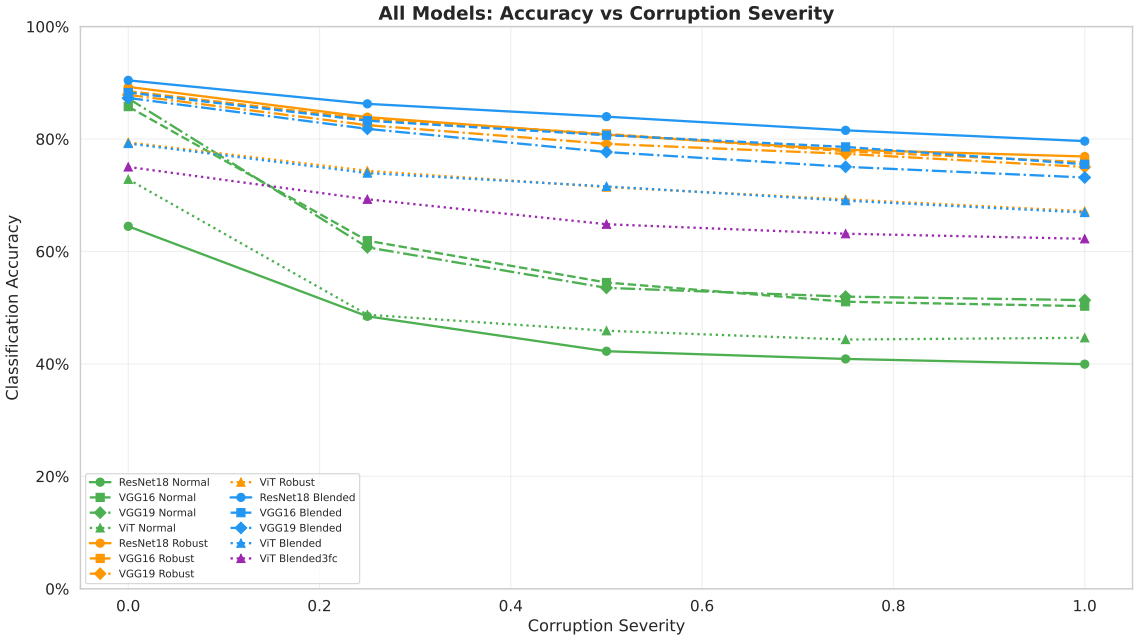
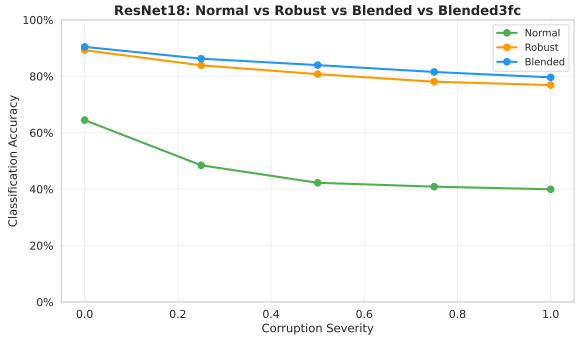
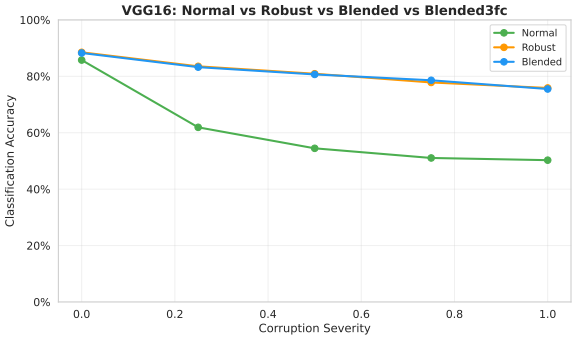


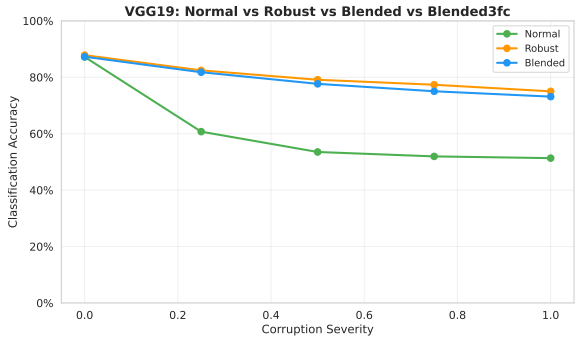
Figure 1: All models: accuracy vs corruption severity.



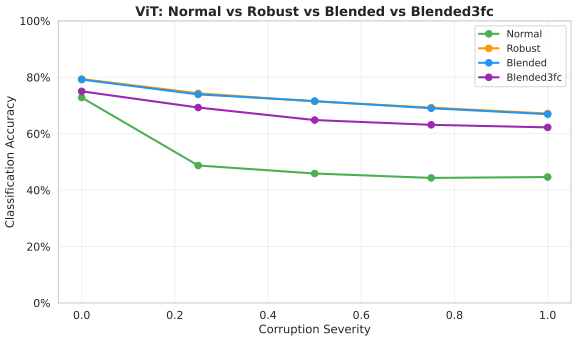
(a) ResNet18



(b) VGG16



(c) VGG19



(d) ViT

Figure 2: Per-backbone comparison: Normal vs Robust vs Blended vs Blended3fc.

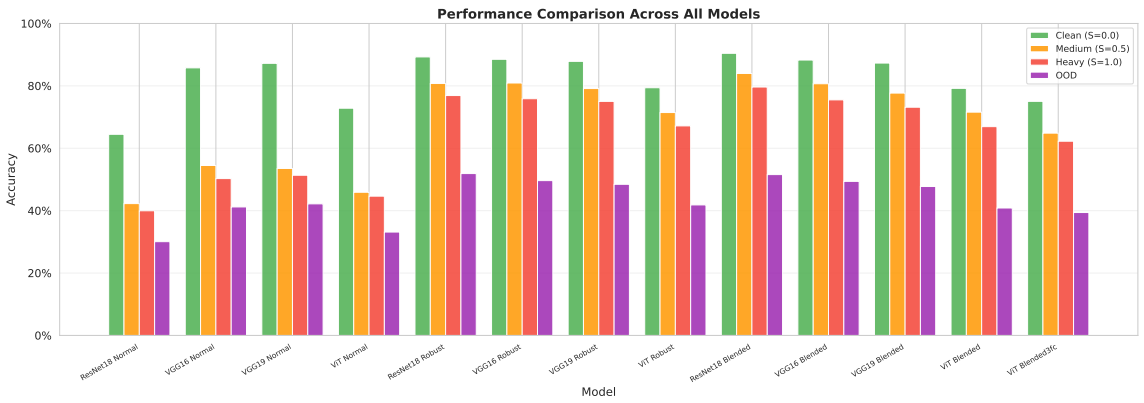


Figure 3: Performance comparison across all models and corruption levels.

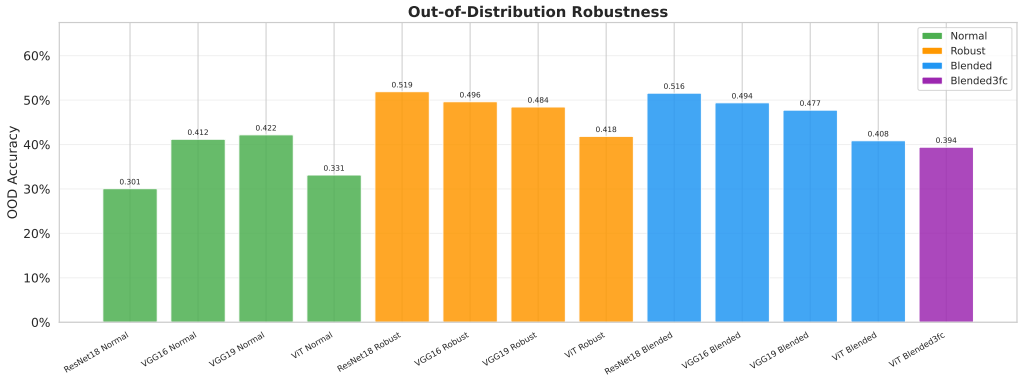


Figure 4: Out-of-distribution robustness comparison.

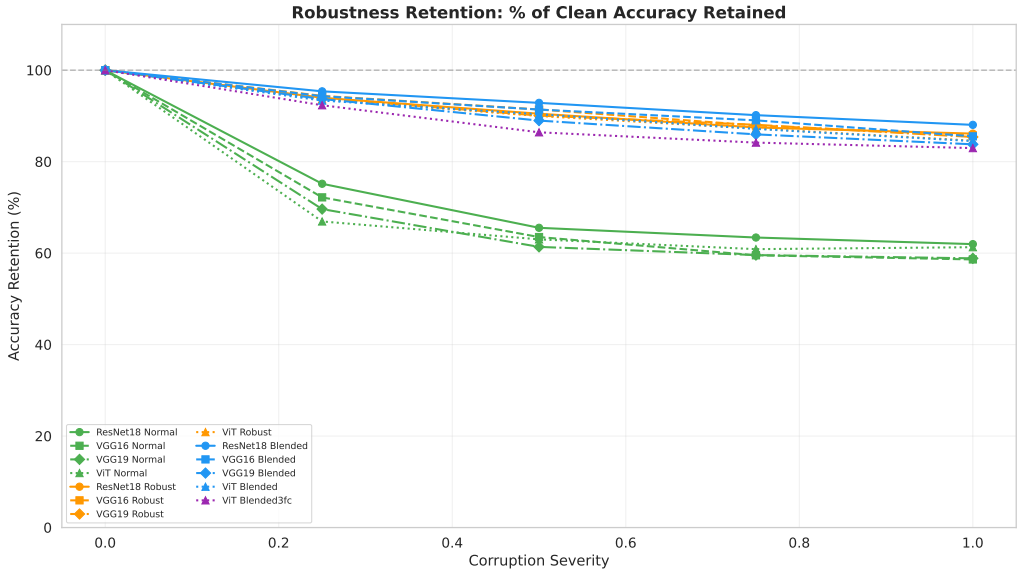


Figure 5: Percentage of clean accuracy retained under corruption.

3 Conclusion

Our experiments demonstrate that predicting augmentation transforms with a minimal auxiliary loss consistently improves model robustness across all tested backbone architectures. The Blended models achieve lower accuracy degradation under corruptions and better out-of-distribution performance compared to both Normal (no augmentation) and Robust (augmented training) baselines.